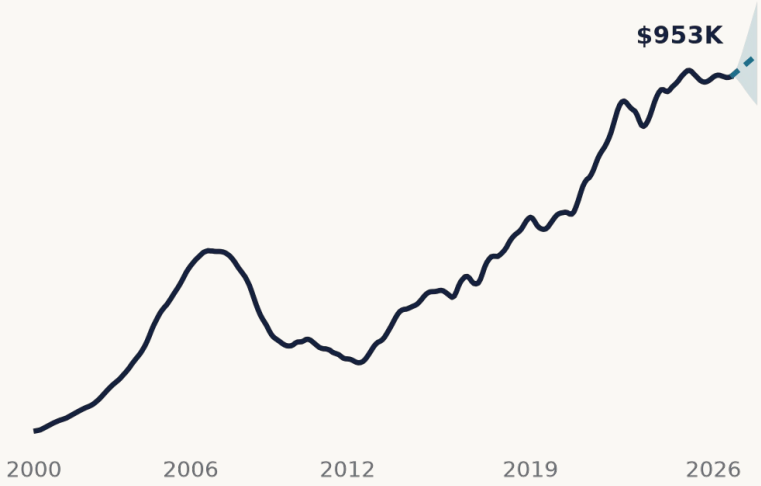
MGMTMSA 437 · FINAL PROJECT · OCTOBER 2026

Forecasting LA Home Values

How well can we forecast LA home values up to a year ahead, and which works best: time-series models or AI?

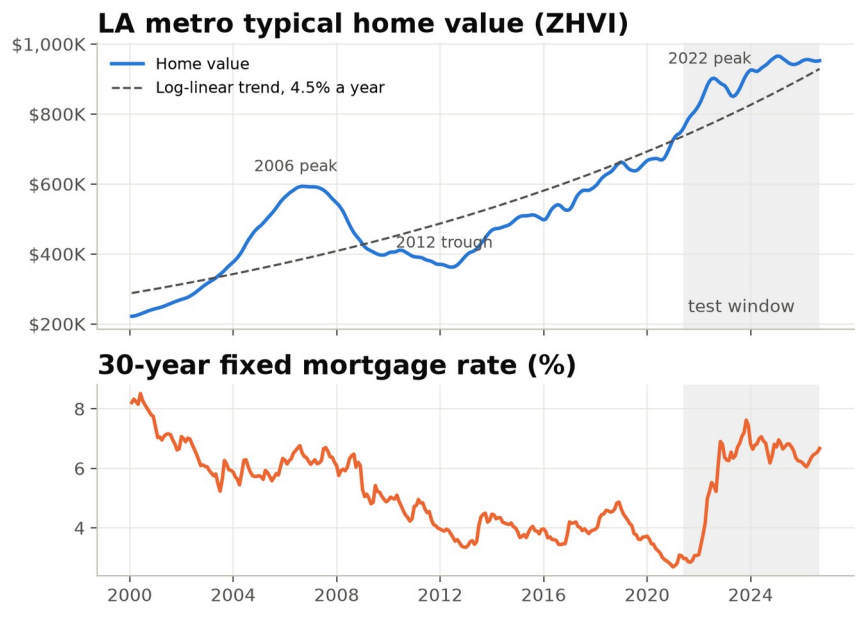
Yang Liu · Jinwoo Roh
Yiu-Hon Lee · Thanapol Doungsaeng

Zillow Home Value Index, LA metro, Jan 2000 to Aug 2026, plus 499 neighborhoods and the 30-year mortgage rate



Solid: actual · Dashed: our forecast to Aug 2027

LA values grew 5.6% a year for 26 years, then stalled when rates hit 7%



$952,601

typical LA-metro home, August 2026

-1.3%

vs the January 2025 peak of $965,634

2.7% → 7.6%

30-year mortgage rate, 2021 low to 2023 high

DATA AND DESIGN

We compared 14 models across 18 forecast dates

DATA

Zillow ZHVI, LA metro
320 months, Jan 2000 to Aug 2026

499 LA neighborhoods
Zillow neighborhood index

Mortgage rate and payroll jobs
FRED and BLS, LA and Orange County

14 MODELS + 3 LLMS

Baseline and classical
Naive drift · ARIMA · Holt · Prophet ×2

ML, deep learning, AI
XGBoost · LightGBM · LSTM · GRU · Chronos

Outside information
ARIMAX ×2 · VAR · factor VAR

Plus 3 chat LLMs
Claude · GPT · Gemini

2 TESTS

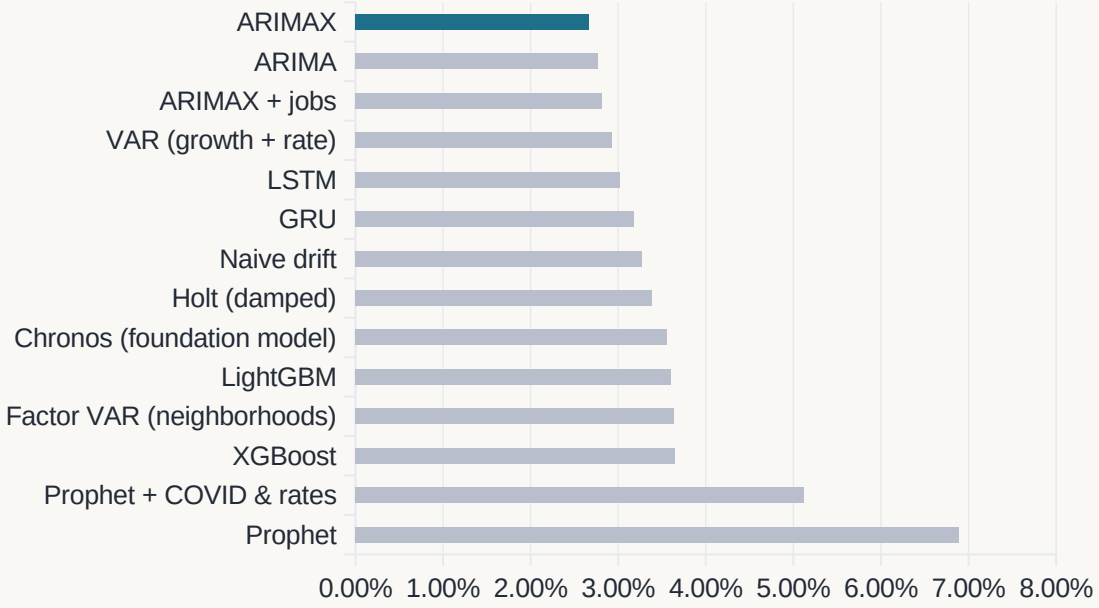
1. Rolling origin (main)
Refit every 3 months, forecast 12 months ahead, 18 times

2. Single blind 80/20
Train to Apr 2021, forecast 64 months in one shot

HOW WE SCORE Error = average absolute % gap between the forecast and Zillow's actual value (MAPE), over 1 to 12 months ahead and all 18 start dates. Lower is better; RMSE in dollars agrees.

RESULT 1 · ROLLING TEST

In the fair test, ARIMA-family models beat machine learning



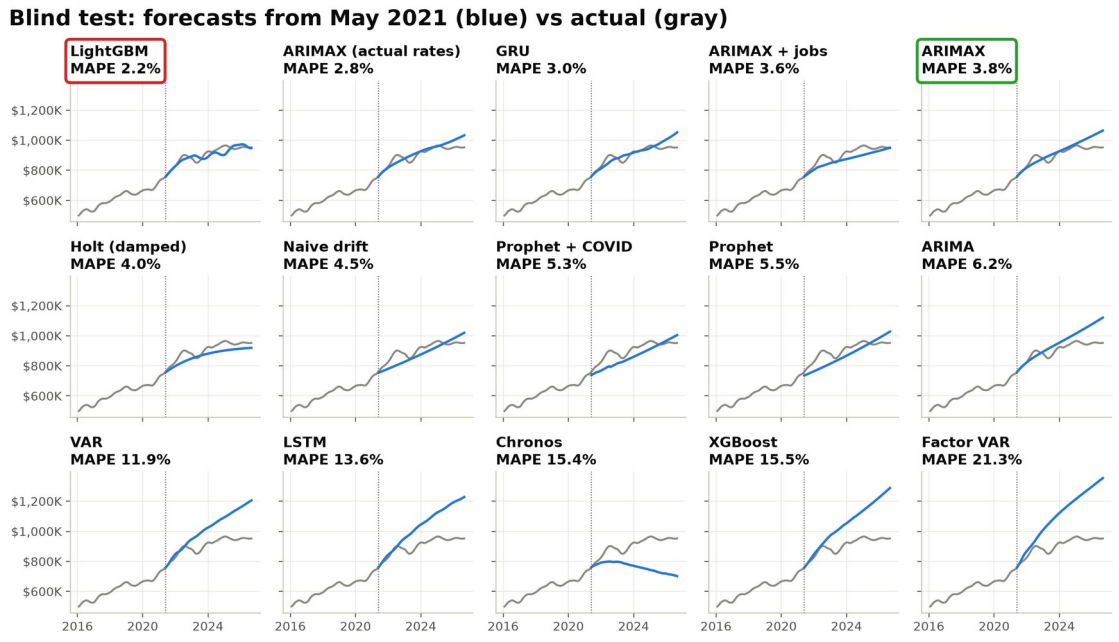
LOWEST ERROR

2.67%

ARIMAX (ARIMA + mortgage rate), average error 1 to 12 months ahead; ARIMA 2.77%. At 12 months: ARIMAX 5.0%, naive drift 4.9%.

RESULT 2 · SINGLE BLIND TEST (ONE WINDOW)

One test window crowned LightGBM; 18 windows ranked it tenth



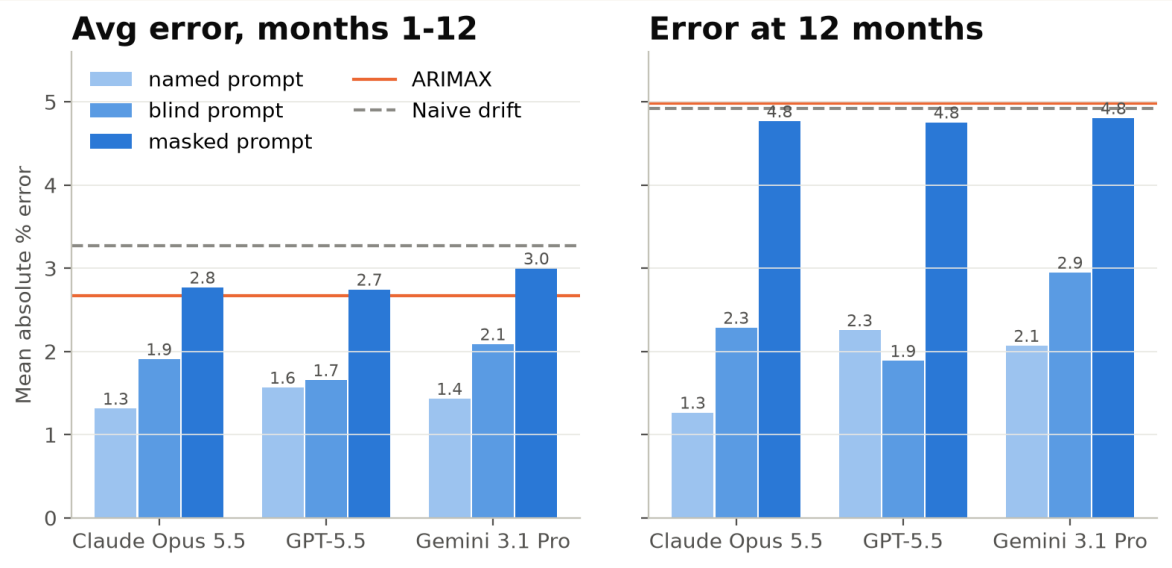
SAME MODEL, TWO TESTS

1st → 10th

LightGBM won the single 64-month test (2.2% error) but ranked 10th across 18 rolling 12-month tests: its lead did not hold.

RESULT 3 · AI AS AN ALTERNATIVE

An alternative tool: AI won by memory, not forecasting



SAME MODEL, DATES HIDDEN

1.3 → 2.8%

Claude's error once it could not tell the place, dates or mortgage rate (ARIMAX 2.67%). All test dates are inside the LLMs' training data.

RESULT 3 · AI AS AN ALTERNATIVE, HOW WE TESTED

The LLMs' edge disappears once we mask the data

NAMED

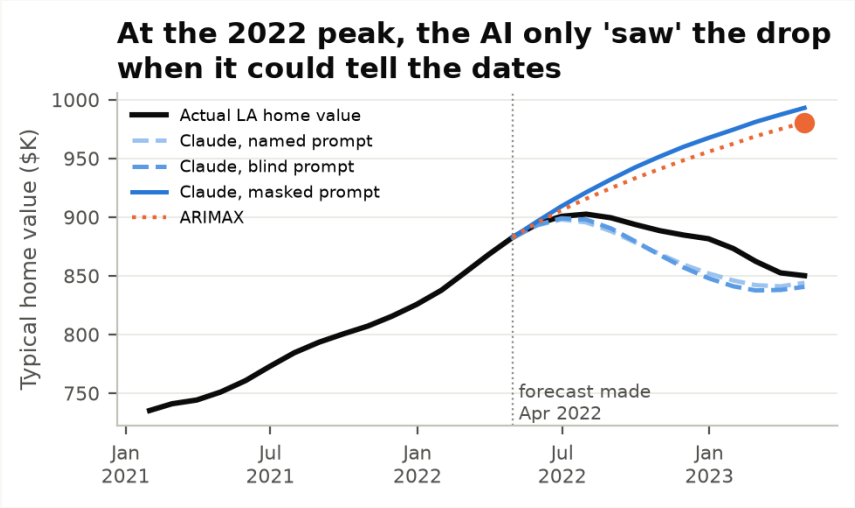
LA, real dates, dollars, mortgage rate
Leaks: it knows the place and the years

BLIND

No place or dates, but the real rate
Leaks: the 2022 rate jump dates the series

MASKED

Last 10 years, rescaled; no place or rate
Fewest clues; period still guessable

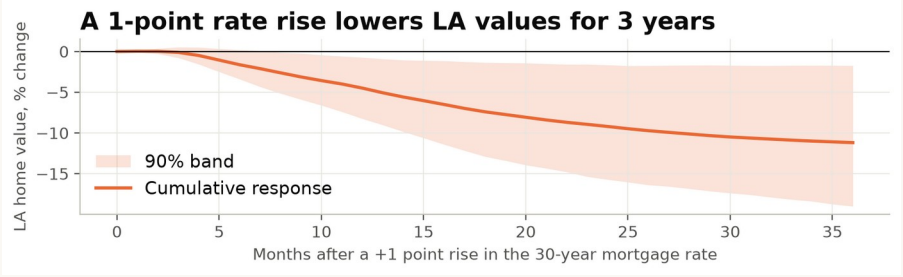


NAME THIS SERIES LA identified: blind 16 of 18 (dates exactly right); masked 0 of 18. The period can still be guessed from the shape.

RESULT 4 · MORTGAGE RATES AND OUTLOOK

Rates act slowly; we forecast +5% by Aug 2027

LA HOUSING VALUE VS MORTGAGE RATES



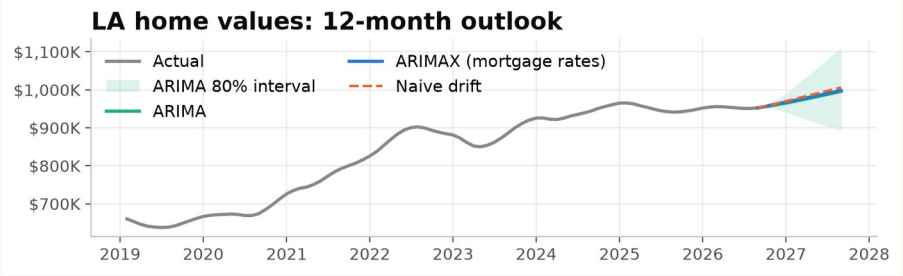
−9.2%

two years after a 1-point rate rise (−4.5% after one)

p = 0.052

rates lead prices (Granger, borderline). ARIMAX: rate p = 0.16, jobs p = 0.29

OUR FORECAST: ABOUT 5% HIGHER BY AUGUST 2027, WITH A WIDE RANGE



$998K (+4.8%)

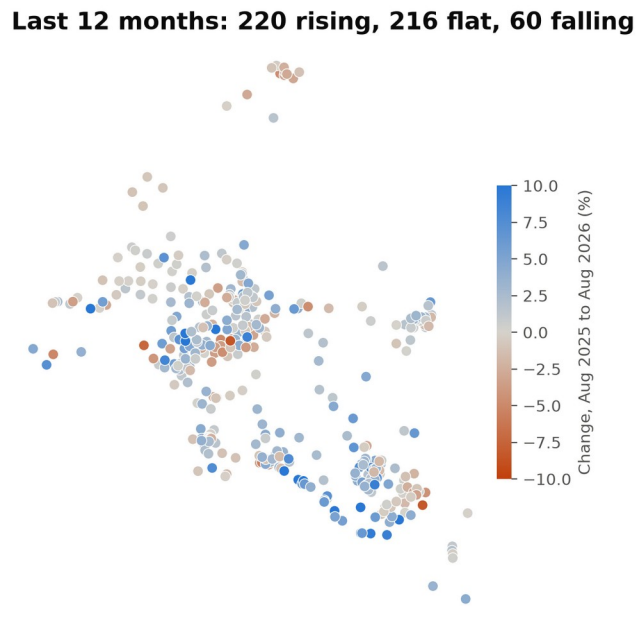
ARIMAX forecast, August 2027

-6% to +17%

ARIMA 80% interval for Aug 2027, for context

NEIGHBORHOODS · LIVE MAP

The metro average hides a split market, neighborhood by neighborhood



220 / 216 / 60

neighborhoods rising >2% / within ±2% / falling >2%, last 12 months (496 with recent data)

152 of 496

still below their 2022 peak, mostly in LA County

−1.2 pts

deeper 2022 drop per doubling of price in LA County (reverses in Orange County)

[Live map: neiltd.github.io/la-housing-forecast](neiltd.github.io/la-housing-forecast)

TAKEAWAYS

What we learned, and what we would do next

01 **Simple beats complex, if you test fairly**

ARIMA models beat ML and Chronos; one blind window would have crowned LightGBM (10th overall).

02 **AI is an alternative only if you test it fairly**

Chatbots beat every model until we hid the place, dates and rate; then no clear edge over ARIMAX.

03 **Mortgage rates matter, slowly; jobs didn't improve our model**

A 1-point rise lowers values about 9% over two years. Turning points stay hard to call.

04 **The metro average hides a split market**

Separate models per neighborhood beat one shared factor (2.8% vs 4.1% error).

NEXT STEPS Test more cities; score the AI live on months after its training cutoff; feed AI altered histories to check whether it forecasts the data or recalls the past.